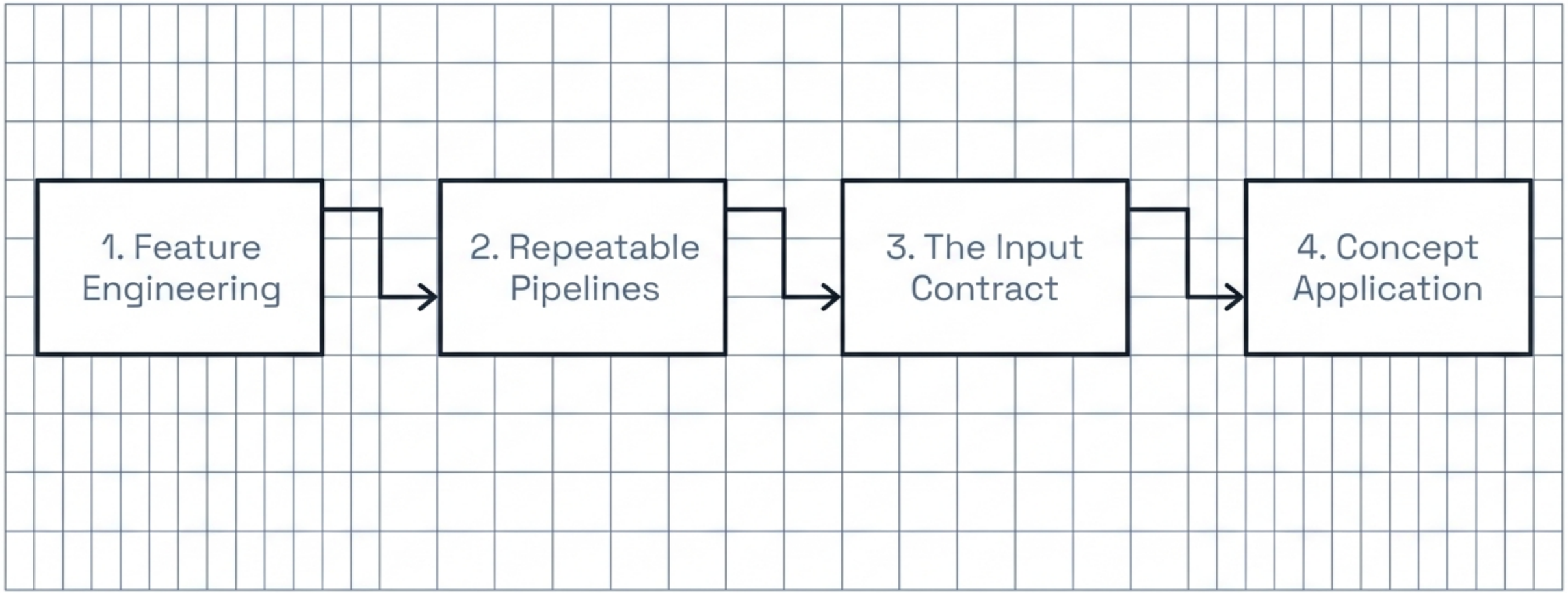
Module 5, Lesson 3

AI 101 FOR MECHANICAL ENGINEERS

Features and repeatable pipelines

Gemini Notebook

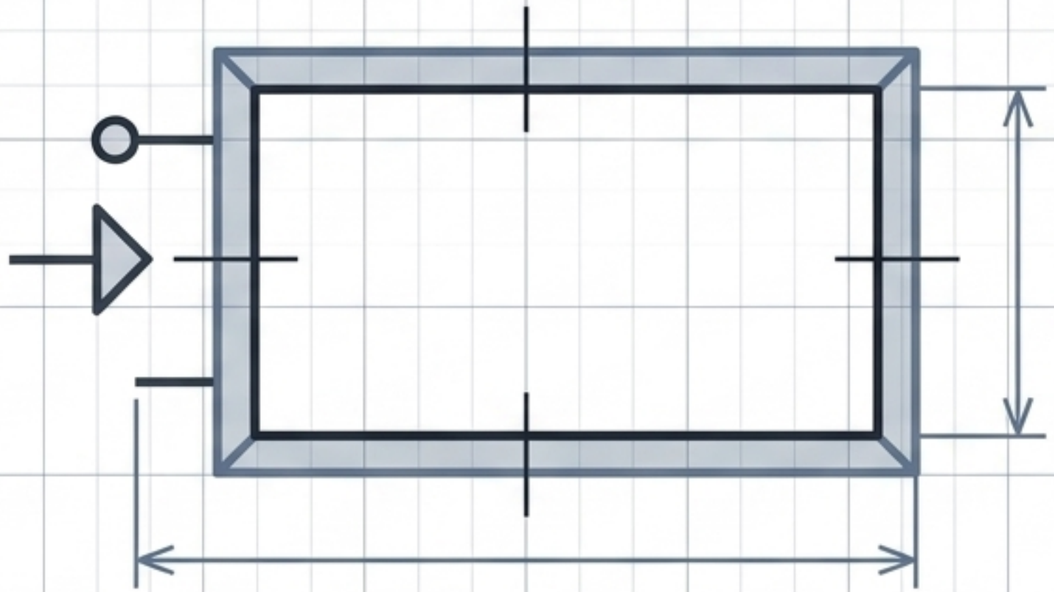
System Structure and Navigation



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1. Feature Engineering

Constructing mathematically and physically rigorous inputs for predictive models.



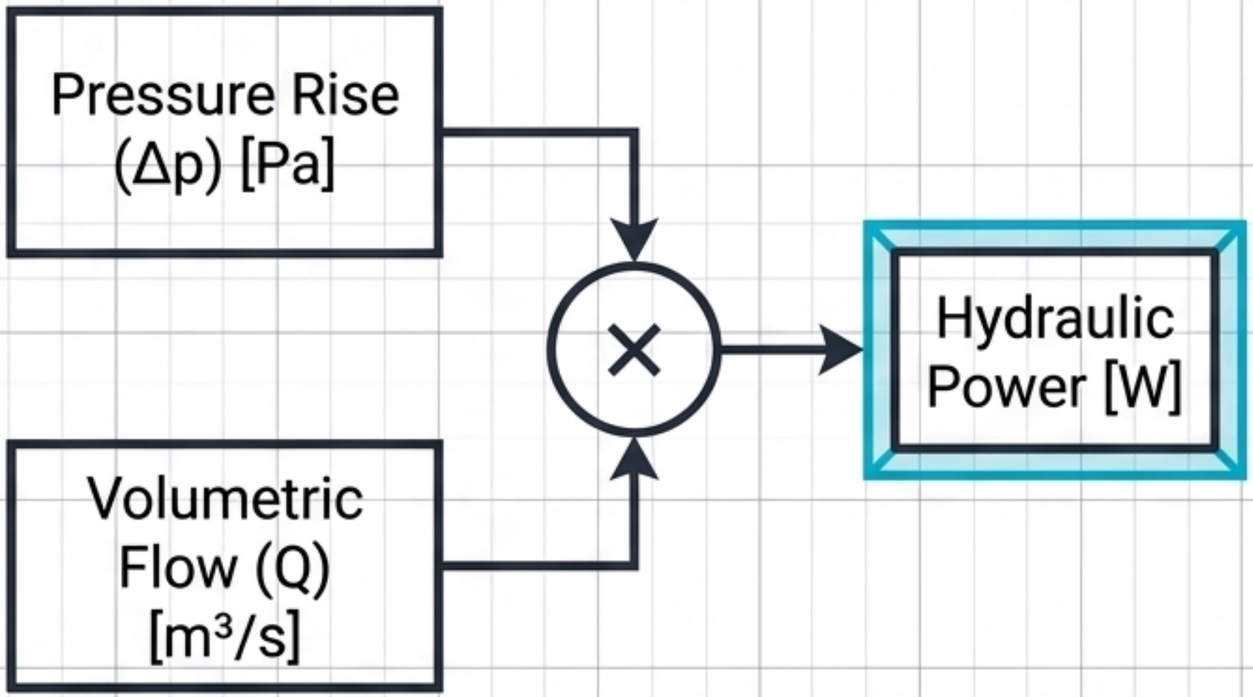
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Deriving Features from Physical Knowledge

Construct meaningful system inputs utilizing domain knowledge.

Features must be strictly available at prediction time.

Example derived feature: Hydraulic power calculation.



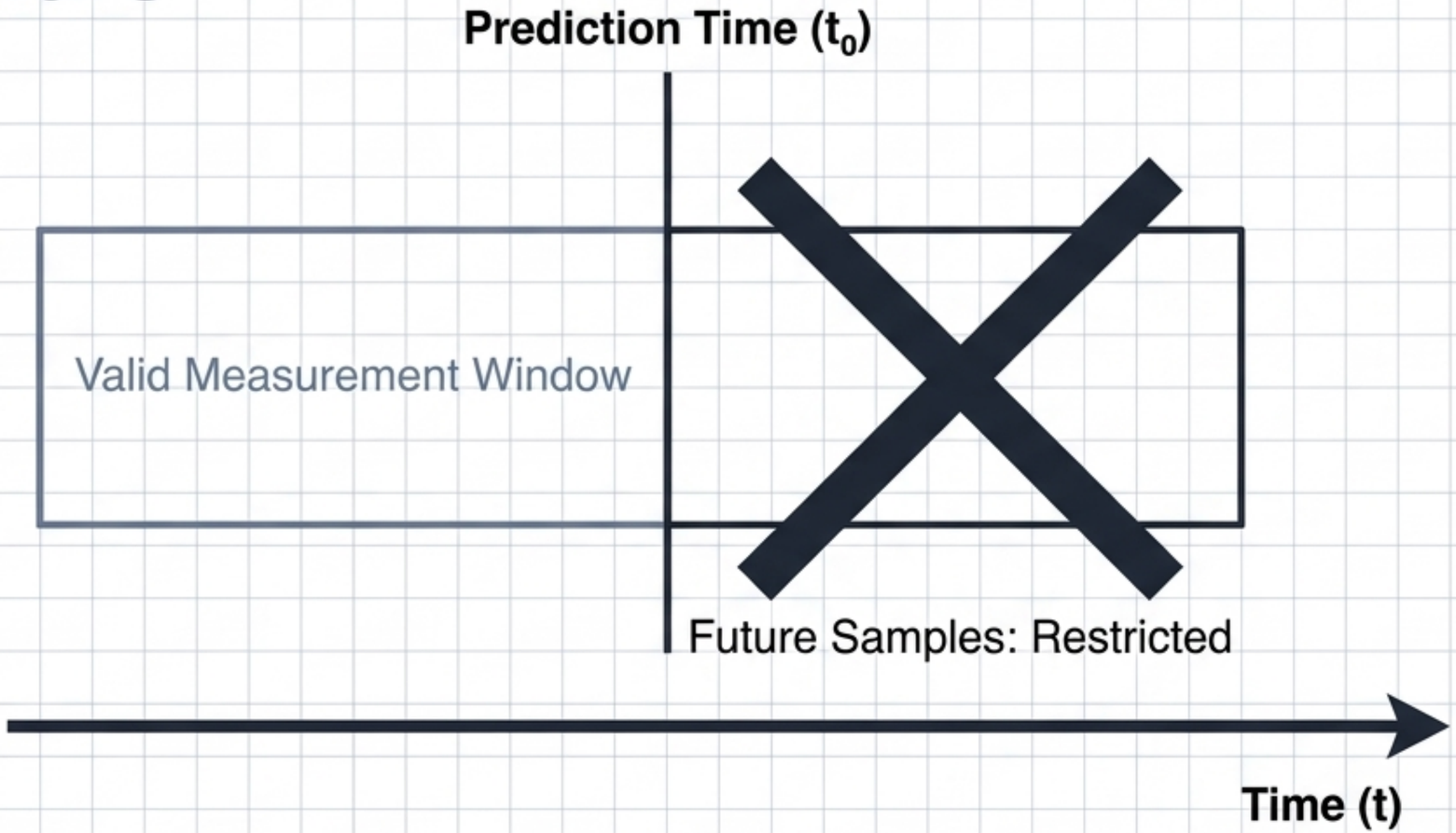
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Time Constraints on Windowed Signals

Windowed signal features process historical sensor data.

Calculations must strictly exclude all future samples.

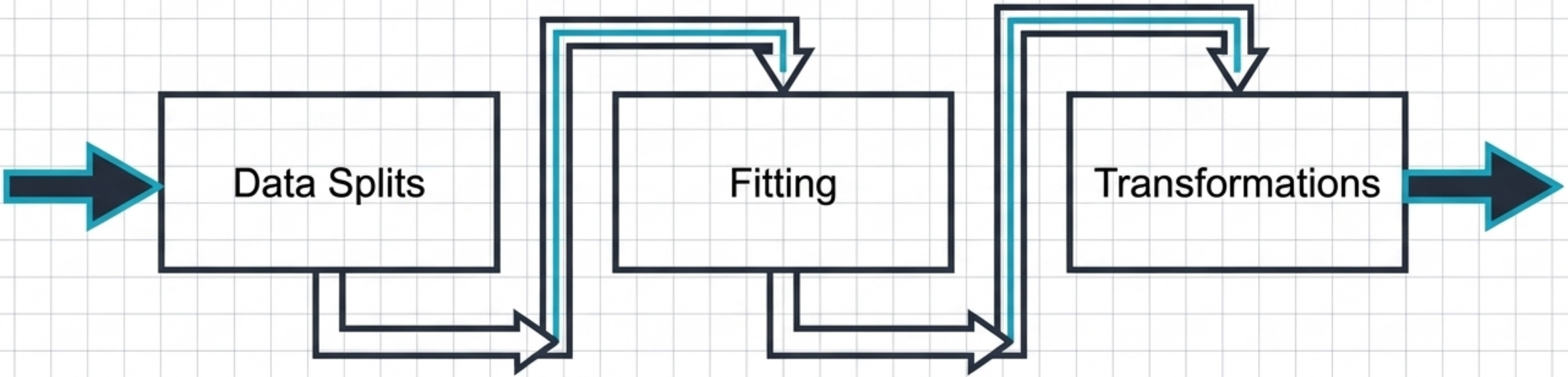
Data leakage physically invalidates the predictive model.



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2. Repeatable Pipelines

Structuring the sequential processing of data splits, fitting, and transformations.



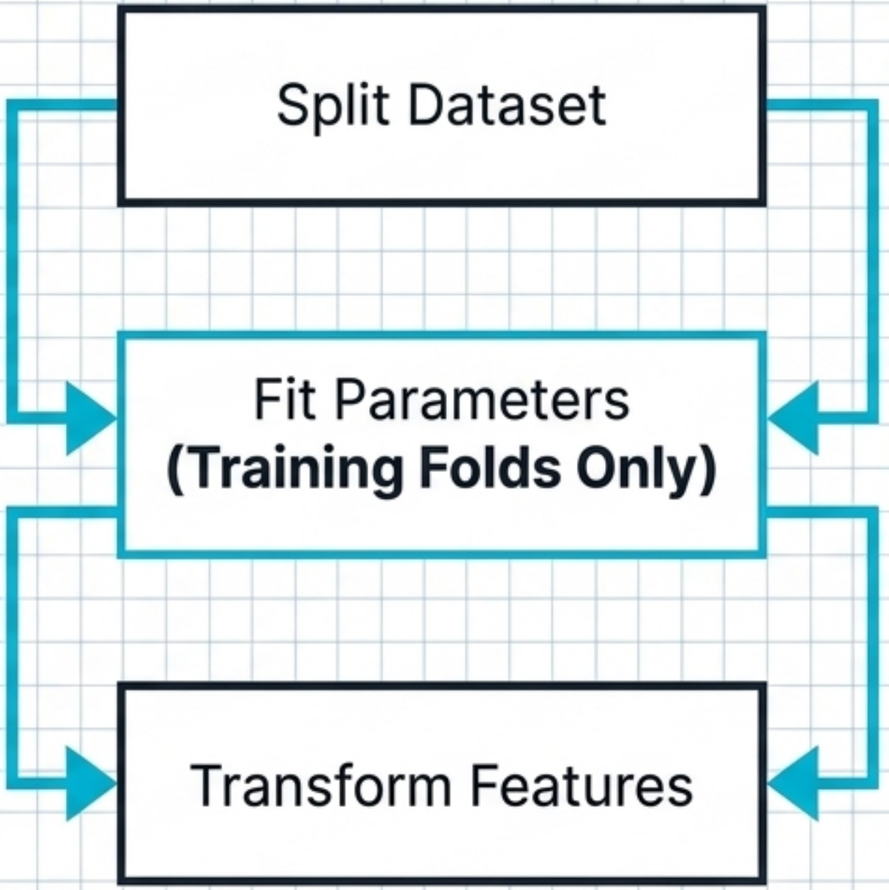
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Preprocessing Pipeline Architecture

A pipeline bundles sequential preprocessing and modelling.

Initialisation requires splitting data before learned operations.

Fit scalers and imputers exclusively within training folds.



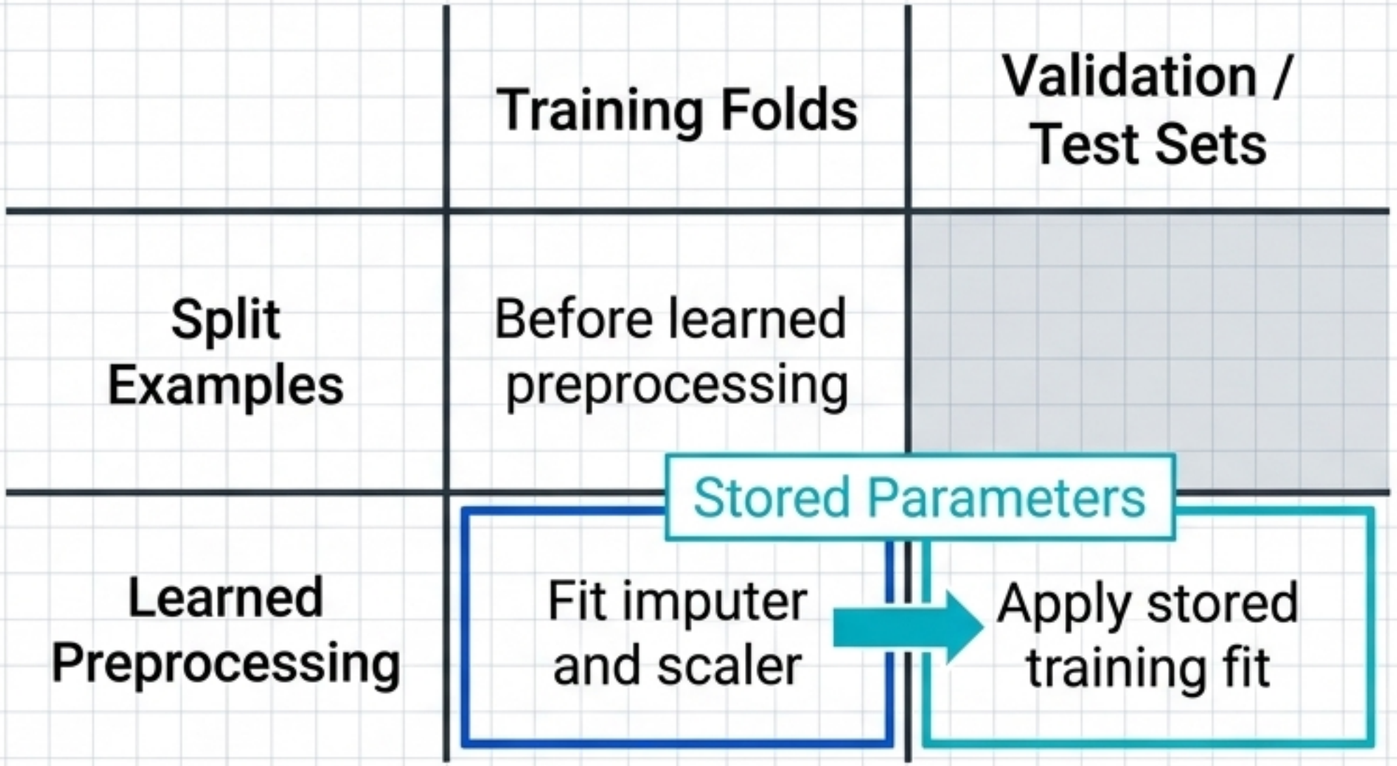
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Fit Versus Application Boundary

Apply fitted transformations to subsequent test inputs.

Do not recalculate statistical parameters on test data.

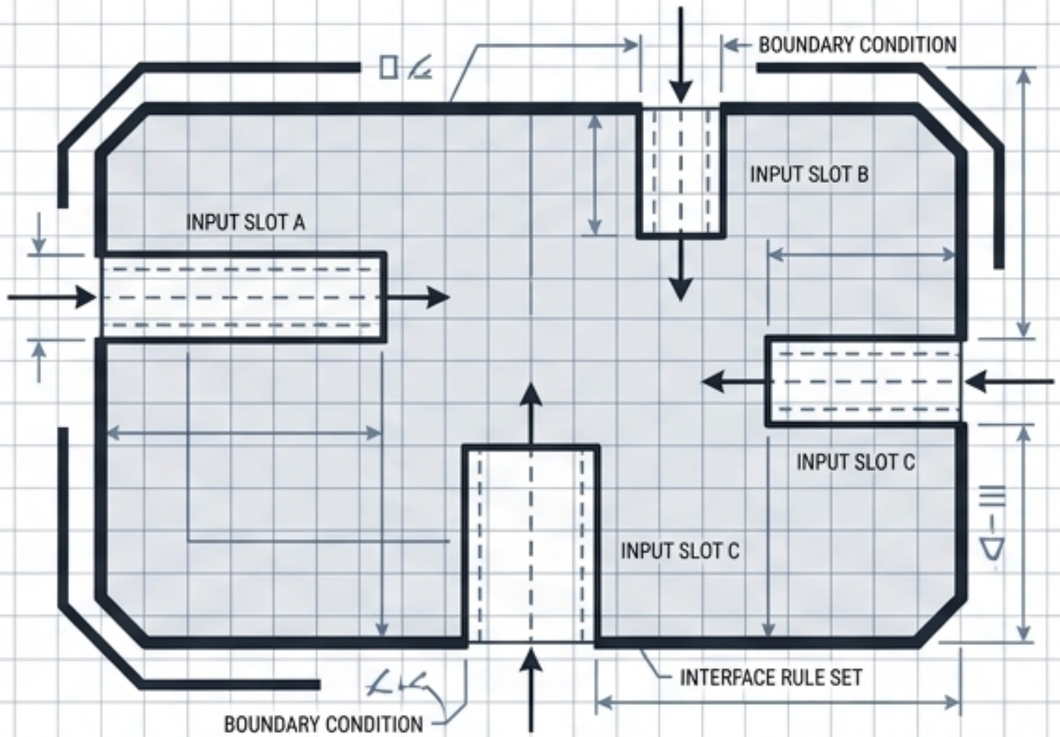
Stored parameters enforce consistency across all sets.



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3. The Input Contract

Defining strict boundary conditions and interface rules for model inputs.



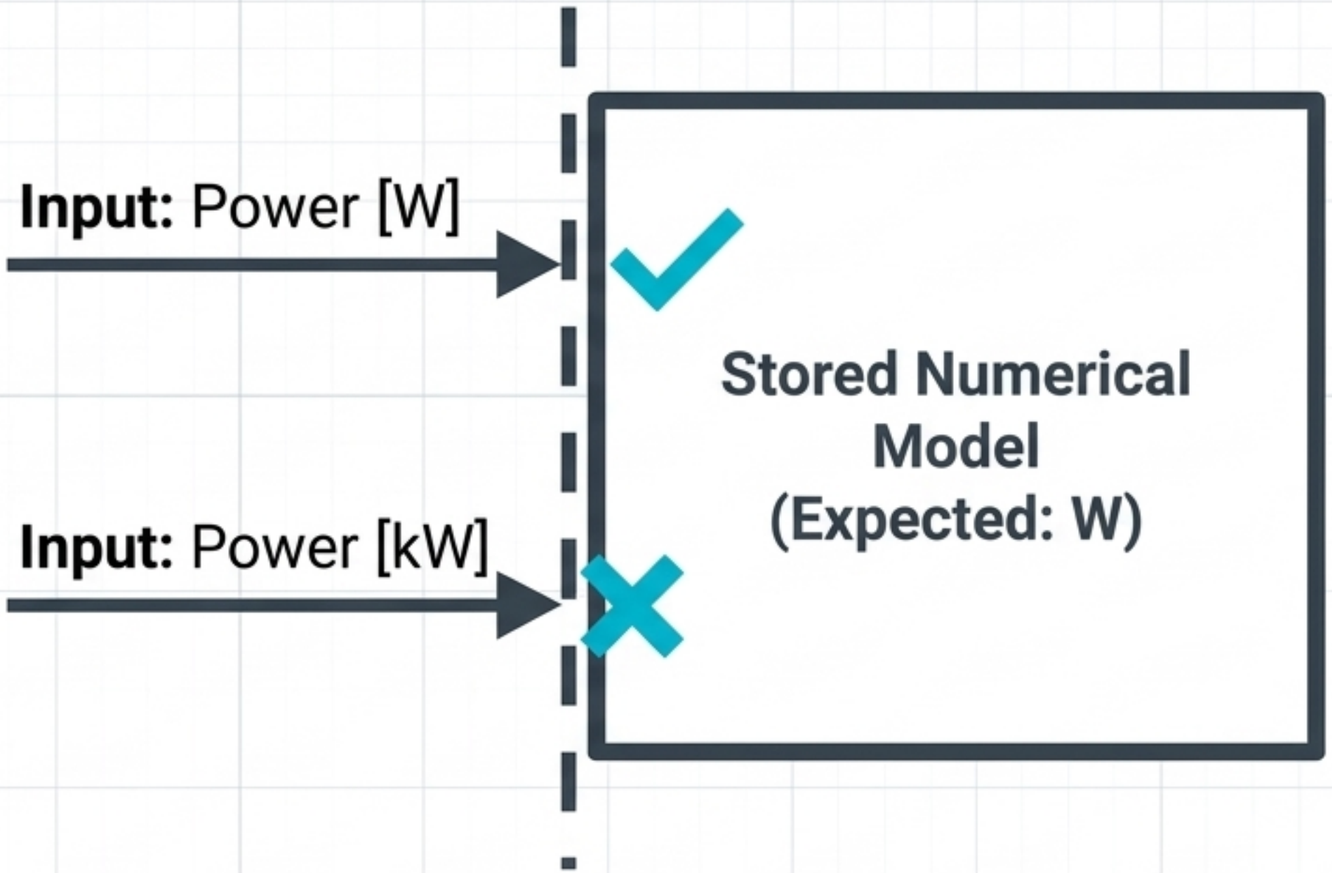
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System Boundary and Unit Validation

Document the expected input unit specifications.

Sensor unit variations compromise stored numerical models.

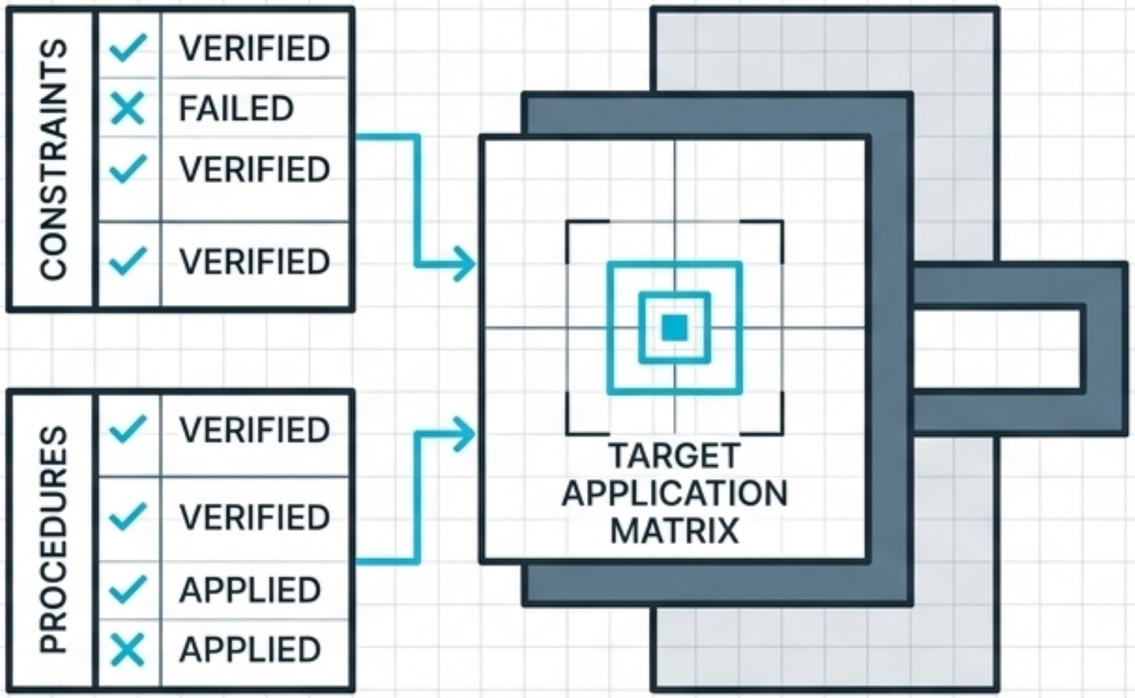
Transformations constitute the logged **evidence trail**.



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4. Concept Application

Verifying constraints and procedures through analytical case application.



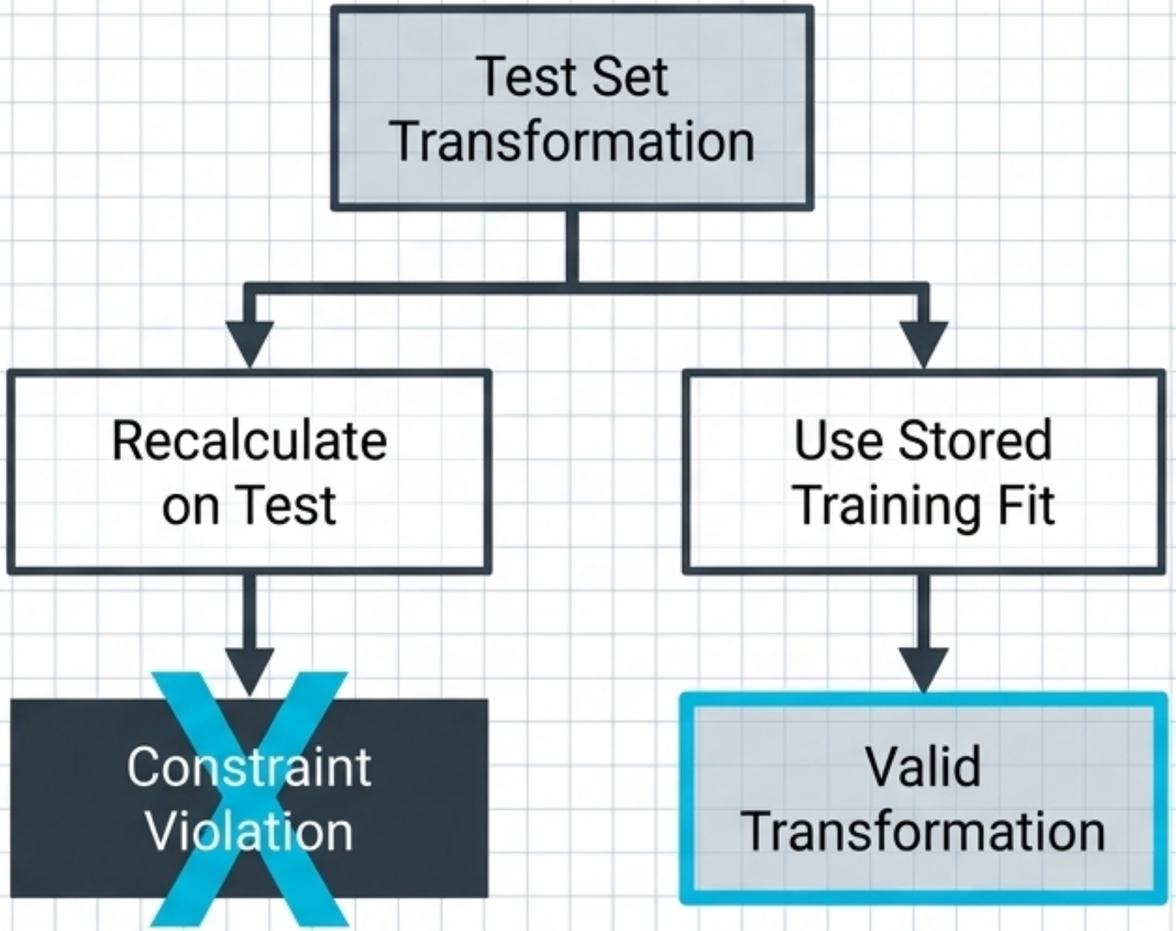
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Application: Scaler Statistics

Problem: Determine scaler statistics for final test sets.

Constraint: Test examples must remain strictly isolated.

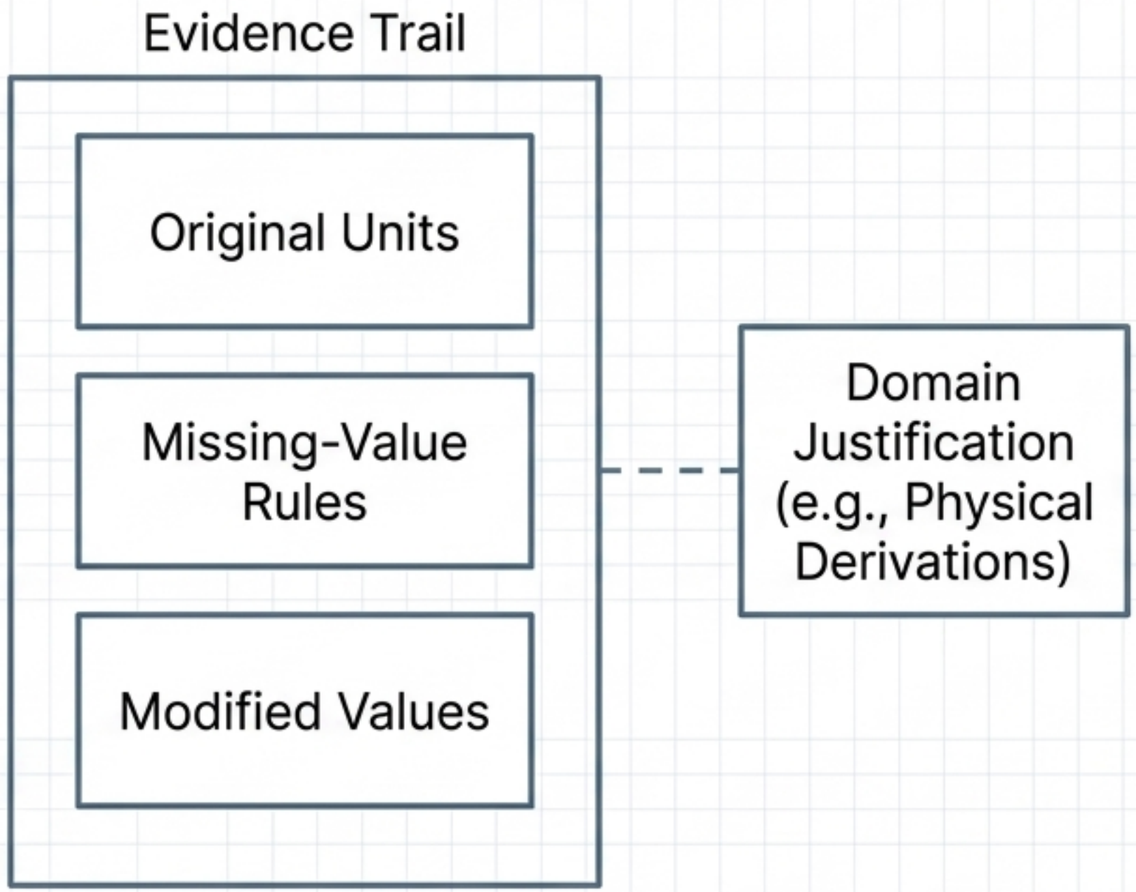
Solution: Apply training-fitted scaler parameters unchanged.



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Application: Evidence and Justification

- Retain original units and missing-value rules.
- Log all specific modified data points.
- Imputation procedures do not recover ground truth.
- Derived features require physical domain justification.



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Key Points

1. Propose meaningful features strictly available at decision time.
2. Bundle preprocessing and modelling into a training-only pipeline.
3. Record the expected input contract to maintain model integrity.
4. Apply fitted transformations to test inputs without refitting.

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